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# Inhaled nitric oxide for the postoperative management of pulmonary hypertension in infants and children with congenital heart disease (Review)

Bizzarro M, Gross I, Barbosa FT

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[Intervention Review]

# Inhaled nitric oxide for the postoperative management of pulmonary hypertension in infants and children with congenital heart disease

Matthew Bizzarro[1], Ian Gross[1], Fabiano T Barbosa[2]

[1]Department of Pediatrics, Yale University School of Medicine, New Haven, CT, USA. [2]Department of Clinical Medicine, Hospital Geral do Estado Professor Osvaldo Brandão Vilela, Maceió, Brazil

**Contact address:** Matthew Bizzarro, Department of Pediatrics, Yale University School of Medicine, 333 Cedar Street WP493, P.O. Box 208064, New Haven, CT, 06520-8064, USA. matthew.bizzarro@yale.edu.

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## ABSTRACT

### Background

Nitric oxide (NO) is a prevalent molecule in humans that is responsible for many physiologic activities including pulmonary vasodilation. An exogenous, inhaled form (iNO) exists that mimics this action without affecting systemic blood pressure. This therapy has been implemented in the treatment of pulmonary hypertension. This review examines the efficacy of iNO in the postoperative management of infants and children with congenital heart disease (CHD). The original review was published in 2005, updated in 2008 and again in 2014.

### Objectives

To compare the effects of postoperative administration of iNO versus placebo or conventional management, or both, on infants and children with CHD and pulmonary hypertension. The primary outcome was mortality. Secondary outcomes included length of hospital stay; neurodevelopmental disability; number of pulmonary hypertensive crises (PHTC); changes in mean pulmonary arterial pressure (MPAP), mean arterial pressure (MAP), and heart rate (HR); changes in oxygenation measured as the ratio of arterial oxygen tension ($PaO_2$) to fraction of inspired oxygen ($FiO_2$); and measurement of maximum methaemoglobin level as a marker of toxicity.

### Search methods

In this updated version we extended the CENTRAL search to 2013, Issue 12 of *The Cochrane Library*, and MEDLINE and EMBASE through to 1 December 2013. The original search was performed in July 2004 and again in November 2007. We included abstracts and all languages.

### Selection criteria

We included randomized and quasi-randomized controlled trials comparing iNO with placebo or conventional management, or both. Trials included only children with CHD requiring surgery complicated by pulmonary hypertension.

### Data collection and analysis

Two authors extracted data. Data were collected on mortality; number of PHTC; changes in MPAP, MAP, HR, and $PaO_2$:$FiO_2$; and maximum methaemoglobin level. Data on long-term mortality, neurodevelopmental disability, and length of hospital stay were unavailable. We performed subgroup analysis by method of control (placebo or conventional management).


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**Main results**

We reran the searches to December 2013 and identified three new studies. These three studies did not fulfil our inclusion criteria. Therefore, no new studies were included in this updated review. In total four randomized trials involving 210 participants were included in this review. We observed no differences in mortality (OR 1.67, 95% CI 0.38 to 7.30; P = 0.50); PHTC (OR 0.80, 95% CI 0.15 to 4.18; P = 0.79); changes in MPAP (treatment effect -2.94 mm Hg, 95% CI -9.28 to 3.40; P = 0.36), MAP (treatment effect -3.55 mm Hg, 95% CI -11.86 to 4.76; P = 0.40), HR (treatment effect 0.02 bpm, 95% CI -8.13 to 8.18; P = 1.00), or $PaO_2$:$FiO_2$ (mean difference 17.18, 95% CI -28.21 to 62.57; P = 0.46). There was a significant increase in the methaemoglobin level (mean difference 0.30%, 95% CI 0.24 to 0.36; P < 0.00001) in patients treated with iNO, although levels did not reach toxicity levels. Data from long-term mortality, neurodevelopmental disability, and length of stay were not available. Two trials had a low risk of bias. Very low quality of the evidence was observed considering grading of the outcomes.

**Authors' conclusions**

We observed no differences with the use of iNO in the outcomes reviewed. No data were available for several clinical outcomes including long-term mortality and neurodevelopmental outcome. We found it difficult to draw valid conclusions given concerns regarding methodologic quality, sample size, and heterogeneity.

## PLAIN LANGUAGE SUMMARY

### Inhaled nitric oxide to treat high blood pressure of the pulmonary vessels after surgery in infants and children with congenital heart disease

**Review question**

We reviewed the evidence on the effectiveness of inhaled nitric oxide for treatment of high blood pressure in the pulmonary vessels in infants and children born with heart disease.

**Background**

Pulmonary hypertension (PH) is an increase in blood pressure in the pulmonary vessels that affects various patient populations and can cause poor health and a significant number of deaths. PH can reduce cardiac function and precipitate a life-threatening crisis. Congenital heart disease (CHD) is a structural deformity of the heart that exists at birth, and often before birth. PH may arise because of increased pulmonary blood flow and persistent raised pulmonary arterial pressures, or following the use of cardiopulmonary bypass during surgical repair. Inhaled nitric oxide (iNO) is a therapy which causes a selective reduction in the high blood pressure of the pulmonary vessels without decreasing blood pressure in the rest of the body and, therefore, may have a treatment benefit in children with heart disease. In this review, we aimed to assess the benefits and harms of using iNO for postoperative management of PH in children with CHD.

**Study characteristics**

The evidence was current to December 2013. We found four randomized clinical trials involving 210 participants aged from one day to 17 years with PH either in the preoperative (one study) or postoperative period (three studies). Control groups received conventional management therapy (two trials) or nitrogen gas as placebo (two trials). Two trials compared changes in systemic and pulmonary arterial blood pressure and heart rate (haemodynamics). The other two trials compared the number of pulmonary hypertensive crises and deaths, with changes in haemodynamic measurements as secondary outcomes.

**Key results**

We found no differences between the groups of patients who received iNO and those that did not. Two trials reported on deaths before discharge, with none occurring in one of the trials. We observed no differences in the number of deaths (two trials); pulmonary hypertensive crises (one trial); changes in mean pulmonary arterial pressure (three trials), arterial pressure (three trials), or heart rate (HR) (three trials); or changes in oxygenation of the blood (one trial). However, no trials reported long-term deaths or neurodevelopmental disability. In addition, no data were available for analysis of length of hospital stay.

Although iNO has been studied as a postsurgical therapy in children with heart disease in order to assist recovery, this review showed no benefits with its use.

**Quality of the evidence**

Two trials had a low risk of bias. The quality of the evidence was, however, very low due to the small number of participants and low event rates. All trials utilized different concentrations of iNO, different durations of administration initiated at different times after the operation, and included patients with diverse congenital heart defects necessitating repair.


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## SUMMARY OF FINDINGS

**Summary of findings for the main comparison. Inhaled nitric oxide compared to placebo or conventional therapy for all infants and children with congenital heart disease having postoperative management of pulmonary hypertension**

**Inhaled nitric oxide compared to placebo or conventional therapy for all infants and children with congenital heart disease having postoperative management of pulmonary hypertension**

**Patient or population:** all infants and children with congenital heart disease having postoperative management of pulmonary hypertension
**Settings:** postoperative
**Intervention:** inhaled nitric oxide
**Comparison:** placebo or conventional therapy

| Outcomes | Illustrative comparative risks* (95% CI) | | Relative effect (95% CI) | No of participants (studies) | Quality of the evidence (GRADE) | Comments |
|---|---|---|---|---|---|---|
| | Assumed risk | Corresponding risk | | | | |
| | Placebo or conventional therapy | Inhaled nitric oxide | | | | |
| **Long-term mortality** | **Study population** | | Not estimable | 0 (0) | See comment | No studies met outcome definition |
| | See comment | See comment | | | | |
| | **Moderate** | | | | | |
| | See comment | See comment | | | | |
| **Mortality prior to discharge** [1] Follow-up: mean 15 days | **Study population** | | **OR 1.67** (0.38 to 7.3) | 162 (2 studies) | ⊕⊝⊝⊝ **very low** [2,3] | |
| | **37 per 1000** | **60 per 1000** (14 to 219) | | | | |
| | **Moderate** | | | | | |
| | **25 per 1000** | **41 per 1000** (10 to 158) | | | | |
| **Difference in MPAP change score (SE) - Conventional Management** [4] Follow-up: mean 5 days | See comment | See comment | | 4 (2 studies) | ⊕⊝⊝⊝ **very low** [4,5] | -1.14 (-8.87 to 6.59) |


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| | | | | | | |
|---|---|---|---|---|---|---|
| **Difference in MPAP change score (SE) - Placebo** [4]<br>Follow-up: mean 5 days | See comment | See comment | Not estimable[4] | 2<br>(1 study) | ⊕⊝⊝⊝<br>**very low** [6,7] | -9.15 (-20.7 to 2.4) |
| **Difference in MAP change score (SE) - Conventional Mangement** [4]<br>Follow-up: mean 5 days | See comment | See comment | | 4<br>(2 studies) | ⊕⊝⊝⊝<br>**very low** [4,5,8] | -1.94 (-11.23 to 7.35) |
| **Difference in MAP change score (SE) - Placebo** [4]<br>Follow-up: mean 5 days | See comment | See comment | Not estimable[4] | 2<br>(1 study) | ⊕⊝⊝⊝<br>**very low** [6,7] | -10 (-28.6 to 8.6) |

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in footnotes. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).
**CI:** Confidence interval; **RR:** Risk ratio; **OR:** Odds ratio

GRADE Working Group grades of evidence
**High quality:** Further research is very unlikely to change our confidence in the estimate of effect.
**Moderate quality:** Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
**Low quality:** Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
**Very low quality:** We are very uncertain about the estimate.

[1] One of the two trials had zero events (Day 2000).
[2] Statistical heterogeneity could not be measured. One of the two trials had zero events (Day 2000).
[3] There was very low precision because of the small numbers and the low rate of events (only 162 patients and 8 events). One of the two trials had zero events (Day 2000).
[4] Pre-specified subgroup analysis.
[5] One of the two trials included was assessed as having a low risk of bias (Day 2000).
[6] Pre-specified subgroup analysis. Study was assessed as having high risk of bias (Russell 1998).
[7] Statistical heterogeneity could not be measured.
[8] There was very low precision because of the small numbers (only 50 patients).


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# BACKGROUND

## Description of the condition

The efficacy of iNO has been extensively investigated and substantiated in the treatment of persistent pulmonary hypertension of the newborn (Clark 2000; Kinsella 1992; NINOS 1997; Roberts 1992; Roberts 1997). The extreme vasoreactivity of the newborn pulmonary vasculature to inhaled nitric oxide (iNO) has raised the question of whether or not iNO would be an effective treatment for pulmonary hypertension in more diverse populations and clinical scenarios (Haworth 1984; Steudel 1999). For example, pulmonary hypertension may arise in patients with congenital heart disease (a structural deformity in the heart existing at birth and often before birth) (CHD) characterized by increased pulmonary blood flow. The increased circulation can potentiate remodeling of the vascular bed resulting in persistence of elevated pulmonary arterial pressure, which accelerates the need for operative intervention and potentially complicates postoperative recovery (Atz 1997; Wessel 1993). Pulmonary hypertension can occur in about 50% of CHD cases (Gorenflo 2010). Endogenous production of nitric oxide is compromised in the early postoperative period and the use of cardiopulmonary bypass during surgical repair can contribute to elevated pulmonary arterial pressures through induction of hypoxaemia, production of microemboli, and excess production of endothelin and thromboxane (Atz 1997). These factors may result in acute elevations in pulmonary vascular resistance that can impede cardiac output and precipitate life-threatening crises (Jones 1981; Wheller 1979). These events occur postoperatively in approximately 7% of patients with CHD, with an associated mortality rate of about 29% (Bando 1996). As a result, researchers and clinicians have investigated the use of iNO in the postoperative management of CHD as a means of reducing pulmonary vascular resistance and averting such complications.

## Description of the intervention

Nitric oxide is a molecule that is produced throughout the human body and has a diversity of physiologic actions. One of interest involves its role as a mediator of vasodilation. Within vascular endothelial cells nitric oxide synthase converts L-arginine into nitric oxide (NO). NO then diffuses into vascular smooth muscle where it binds to soluble guanylate cyclase to convert guanosine triphosphate into cyclic guanosine monophosphate (cGMP). The cGMP then mediates vascular smooth muscle relaxation (Steudel 1999). Exogenous NO in the form of iNO has a similar mechanism of action on the pulmonary vascular bed. After initiating vascular smooth muscle relaxation the molecule diffuses into the bloodstream where it binds with iron in haemoglobin to form methaemoglobin and is inactivated (Steudel 1999). Therefore, the molecule exerts virtually no vasodilatory effect on the systemic circulation. It is this unique property of iNO that has led to the investigation of its use in a myriad of paediatric and adult diseases complicated by pulmonary hypertension (Nelin 1998).

## How the intervention might work

In 1994, Journois et al studied 17 infants with CHD and critical postoperative pulmonary arterial hypertension that was unresponsive to conventional therapies. Initiation of 20 parts per million (ppm) of iNO resulted in a decrease in mean pulmonary arterial pressure (MPAP) in all patients without significant changes in systemic arterial pressure (Journois 1994). Similar findings were published by Miller et al in a case series of 10 infants with CHD who were administered low dose (2, 10, 20 ppm) iNO in the postoperative period. In infants with a high ratio of pulmonary to systemic arterial pressure there was a significant reduction in mean pulmonary vascular resistance with a subsequent rise in cardiac output (Miller 1994). A 1995 case series by Beghetti et al also demonstrated a reduction in pulmonary artery pressure and an increase in arterial oxygen saturation without significant alteration in systemic vascular resistance with the use of continuous iNO in the postoperative period (Beghetti 1995). Around the same time, Curran et al demonstrated a minimal benefit of iNO in patients who had undergone atrioventricular canal repair and a significant reduction in pulmonary artery pressure in those with pulmonary hypertension refractory to conventional therapies (that is hyperventilation, hyperoxygenation, and intravenous inotropic support) (Curran 1995).

## Why it is important to do this review

As a result of the findings described above (Beghetti 1995; Curran 1995; Journois 1994; Miller 1994), several investigators undertook randomized controlled trials (RCTs) to more effectively evaluate the use of iNO in the treatment of pulmonary hypertension in the immediate postoperative period for paediatric patients with CHD. There is a clinical guideline recommending iNO for neonates with acute hypoxic respiratory failure (DiBlasi 2010). Although iNO has been recommended to treat neonates with acute hypoxaemic respiratory failure complicated by pulmonary artery hypertension, its use in other clinical scenarios and patient populations, including infants and children with CHD complicated by pulmonary hypertension, remains uncertain so this systematic review is relevant to answer the research question: what are the effects of postoperative iNO versus placebo or conventional management, or both, on infants and children with CHD?

# OBJECTIVES

To compare the effects of postoperative administration of iNO versus placebo or conventional management, or both, on infants and children with CHD and pulmonary hypertension. We evaluated:

1. long-term mortality;
2. mortality prior to hospital discharge;
3. length of intensive care unit or hospital stay;
4. neurodevelopmental disability (a disorder of brain functionthat affects emotion, learning ability, and memory);
5. pulmonary hypertensive crises (PHTC);
6. MPAP, mean arterial (systemic) pressure (MAP), and heart rate (HR);
7. arterial oxygenation (the amount of oxygen in arterial blood) or saturation (non-invasive measurement of oxygen in arterial blood), or both;
8. methaemoglobin level (an abnormal form of haemoglobin that cannot bind oxygen).

We performed subgroup analysis to observe the effects of iNO by method of control (for example placebo versus use of conventional management).


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# METHODS

## Criteria for considering studies for this review

### Types of studies

We evaluated randomized controlled trials (RCTs) and quasi-randomized controlled trials that compared the use of iNO with placebo or conventional management strategies, or both, and included at least one outcome of interest. We considered both parallel and cross-over design trials for inclusion.

### Types of participants

We included infants and children aged one day to 14 years with CHD requiring surgical repair and with evidence of pulmonary hypertension in the pre- or postoperative period. We excluded preterm neonates.

### Types of interventions

We evaluated the use of iNO at various concentrations ranging from five to 80 ppm when given in the postoperative period.

We evaluated placebo, administered as nitrogen gas ($N_2$), as a control.

Conventional management included hyperventilation to induce respiratory alkalosis, use of sodium bicarbonate to induce metabolic alkalosis, use of intravenous inotropic and vasodilatory agents, and use of intravenous sedatives and neuromuscular blockade in an attempt to improve pulmonary blood flow (Whittsett 1999).

### Types of outcome measures

#### *Primary outcomes*

1. Long-term mortality
2. Mortality prior to discharge

#### *Secondary outcomes*

1. Length of stay in intensive care unit or hospital
2. Incidence and severity of neurodevelopmental disability
3. Incidence of PHTC
4. Change in MPAP
5. Change in MAP
6. Change in HR
7. Change in ratio of arterial oxygen tension ($PaO_2$) to fraction of inspired oxygen ($FiO_2$)
8. Maximum methaemoglobin level

## Search methods for identification of studies

### Electronic searches

In our original review we searched the Cochrane Central Register of Controlled Trials (CENTRAL) (*The Cochrane Library* 2004, Issue 3), MEDLINE (January 1966 to 1 July 2004) and EMBASE (January 1980 to 1 July 2004) (Bizzarro 2005). For our first update, the search was extended from 1 July 2004 through to 1 November 2007.

In this latest updated version we searched CENTRAL through to 2013, Issue 12 of *The Cochrane Library*, and MEDLINE and EMBASE through to 1 December 2013. We used filters to identify RCTs in MEDLINE (Dickersin 1994) and EMBASE (Lefebvre 1996). We limited the searches to studies in humans. We did not apply any language restrictions during the literature search or for trial inclusion. We included abstracts in this search and handsearched conference abstracts for relevant studies. We retrieved all eligible studies in full text.

The strategy for CENTRAL is in Appendix 1.
The strategy for MEDLINE is in Appendix 2.
The strategy for EMBASE in Appendix 3.

### Searching other resources

We searched the reference lists of identified review articles and all included studies in an attempt to identify other potentially eligible studies. We undertook personal communication with primary authors to identify unpublished data, when necessary.

## Data collection and analysis

### Selection of studies

We utilized the standardized methods for conducting a systematic review as described by The Cochrane Collaboration in the *Cochrane Handbook for Systematic Reviews of Interventions* 5.1.0 (Higgins 2011). Two authors independently read the RCTs identified through searching electronic databases and handsearching of journals. Using the inclusion and exclusion criteria defined above, the authors reached consensus as to which studies were included in the review.

### Data extraction and management

Two authors (MB and IG) independently extracted and collected data on a paper form (Appendix 4). We contacted the authors of trials, when necessary, to obtain raw data not presented in the trial report but necessary for the analysis. We resolved any discrepancies in a consensus meeting.

### Assessment of risk of bias in included studies

Two authors (MB and IG) independently assessed the methodologic quality of each included trial following the guidelines described by The Cochrane Collaboration in the *Cochrane Handbook for Systematic Reviews of Interventions* 5.1.0 (Higgins 2011). A copy of the form that we used to do this assessment is in Appendix 4. We resolved any discrepancies in this assessment by a consensus meeting. No third-party intervention was required. We used the 'risk of bias' tool in RevMan 5.2 (RevMan 5.2). We assessed the risk of bias (when there is an error in the conduct of the trial that could lead to a misleading result) based on the information presented in the papers. We did not contact authors for clarification. We used the following as the assessment criteria.

#### *1. Random sequence generation*

Low: sequence generation was reported using computer generated random numbers, codes or sealed envelopes. We judged other processes, such as tossing of a coin, as adequate if the whole sequence was generated prior to the start of the trial and if it was performed by a person not otherwise involved in patient recruitment.

High: a system was used which was generated by a non-random approach, e.g. dates, names or identification numbers.


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Unclear: insufficient information was available to consider 'low' or 'high' risk of bias.

***2. Allocation concealment***

Low: if the process that was used prevented patient recruiters, investigators, and participants from knowing the intervention allocation of the next participant to be enrolled in the study.

High: if the allocation method allowed the patient recruiters, investigators, or participants to know the treatment allocation of the next participant to be enrolled in the study.

Unclear: insufficient information was available to consider 'low' or 'high' risk of bias.

***3. Blinding of participants and personnel***

Low: if participants and personnel were reported as blinded.

High: if participants and personnel were not reported as blinded.

Unclear: insufficient information was available to consider 'low' or 'high' risk of bias.

***4. Blinding of outcome assessment***

Low: if outcome assessors were reported as blinded.

High: if outcome assessors were not reported as blinded.

Unclear: insufficient information was available to consider 'low' or 'high' risk of bias.

***5. Incomplete outcome data***

Low: numbers of withdrawals or exclusions per group, with reasons, were provided or if the authors reported no withdrawals or exclusions.

High: numbers of withdrawals or exclusions per group, with reasons, were not provided or if authors did not report reasons for withdrawals or exclusions per group when they were evident.

Unclear: insufficient information was available to consider 'low' or 'high' risk of bias.

***6. Selective reporting***

Low: if all outcomes were reported.

High: if outcomes were measured but not reported or planned and not measured.

Unclear: insufficient information was available to consider 'low' or 'high' risk of bias.

***7. Other bias***

Low: if the study appeared to be free of other sources of bias not described in other sections of this table.

High: if the study appeared to be fraudulent or had some other problem not described in other sections of this table.

Unclear: insufficient information was available to consider 'low' or 'high' risk of bias.

**Measures of treatment effect**

With respect to data analysis, we employed the standardized methods of the Cochrane Neonatal Review Group (CNRG 2004). These included, but were not limited to:

1. considering which types of study design were appropriate for the review and deciding how to address each type;
2. determining the likely nature of the outcome data (that is, dichotomous, continuous, etc);
3. considering whether it was possible to specify in advance what treatment effect measures were used;
4. deciding how to identify statistical heterogeneity;
5. deciding whether to use random-effects meta-analyses, fixed-effect meta-analyses, or both methods;
6. considering how to assess clinical and methodological diversity (heterogeneity) and whether (and how) to incorporate these into the analysis strategy;
7. deciding how to assess and address the quality of included studies in the analysis; and
8. deciding whether (and how) to seek evidence of possible publication or reporting biases.

We identified pertinent data from trials of both parallel (Day 2000; Miller 2000; Russell 1998) and cross-over (Morris 2000) design. When the outcome of interest included only dichotomous data from parallel trials, we calculated an odds ratio (OR) with 95% confidence interval (CI) using a fixed-effect model. When the outcome of interest included only continuous data from parallel trials, we calculated a mean difference (MD) with 95% CI using a fixed-effect model. When the outcome of interest included data from both parallel and cross-over trials, we employed the generic inverse variance method to analyse these combinations of data. Before any of these methods could be utilized, we collected and assessed individual patient data. We collected baseline measurements of MPAP, MAP, HR, and the ratio of $PaO_2$:$FiO_2$ for each individual patient and compared them with measurements taken at the completion of therapy. We then calculated a mean change score with standard deviation for both the treatment and control groups. We calculated a reduction in any of the above outcomes from baseline as a negative change score, whereas we calculated an increase from baseline as a positive change score. Individual patient data were available from Russell 1998 and Morris 2000; and we obtained data direct from the authors of Day 2000. When necessary, we calculated MPAP from available individual patient pulmonary systolic and diastolic pressures. In Russell 1998, individual patient data were presented as the baseline MPAP of each patient and the subsequent per cent change in MPAP from baseline after initiation of therapy. Using the per cent change from the initial MPAP, the final MPAP and change score could be calculated for each patient. For the change in HR and MAP in Russell 1998, we calculated a change score from the available baseline and post-treatment mean values. To calculate the standard deviation (SD) we utilized the methods described by The Cochrane Collaboration in the *Cochrane Handbook for Systematic Reviews of Interventions* 5.1.0 (Section 7.7.3.2) (Higgins 2011). First, we imputed correlation coefficients for both the treatment and control groups from Day 2000. We chose this study because it was most similar to Russell 1998 (the Morris 2000 trial being of cross-over design). Next, we calculated the SD for


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both the control and treatment group utilizing the corresponding correlation coefficient and the available data.

### Unit of analysis issues

We did not have any unit of analysis issues.

### Dealing with missing data

We contacted the trial authors to clarify missing or unclear data when necessary.

### Assessment of heterogeneity

We performed testing for heterogeneity of treatment effect using a $Chi^2$ test. We also employed the $I^2$ statistic and $Chi^2$ test to measure the degree of inconsistency among results. We considered data highly suspicious if we identified an $I^2$ value greater than 50% and, under these circumstances, data were reanalysed using a random-effects model.

### Assessment of reporting biases

We planned to assess publication bias using a funnel plot (Higgins 2011) if more than nine studies were included in the meta-analysis. The funnel plot has an inverted funnel shape if publication bias is absent; funnel plots are examined visually and through Egger's test to test asymmetry.

### Data synthesis

In order to combine and analyse data from both parallel and cross-over trials, we first calculated treatment effects and 95% CIs for both the continuous and dichotomous data. For continuous data, we calculated an MD with 95% CI using a fixed-effect model for each outcome described. For dichotomous data, we calculated an OR with 95% CI using a fixed-effect model. Next, we calculated a standard error (SE) for each outcome of interest using the 95% CI and a value from the t-distribution as described in the *Cochrane Handbook for Systematic Reviews of Interventions* 5.1.0 (Section 7.7.7.2) (Higgins 2011). Now, with the available treatment effect and SE for each outcome of interest, we could enter data into the generic inverse variance model for analysis. We entered dichotomous data presented in the form of an OR as logarithms as described in the *Cochrane Handbook for Systematic Reviews of Interventions* 5.1.0 (Section 7.7.2) (Higgins 2011). Because the generic inverse variance model output omits the number of data points contributed by each trial, we have included this information in the 'Additional tables' section of the review (see Table 1; Table 2; Table 3;). We performed the analysis in RevMan 5.2 (RevMan 5.2).

### Subgroup analysis and investigation of heterogeneity

We planned to do subgroup analysis based on method of control as described above in the Objectives (placebo or conventional management).

We estimated statistical heterogeneity through the $Chi^2$ test and $I^2$ statistic. Potential sources of heterogeneity among these studies included differential exposure intensity (for example dosing of iNO and length of treatment), differential treatment of control groups (for example placebo versus conventional management), and host factors (for example age, type of underlying cardiac disease). We planned to explore some of these via subgroup analysis.

### Sensitivity analysis

We performed sensitivity analyses, when possible, by excluding trials of lower methodologic quality and reanalysing the data.

## RESULTS

### Description of studies

The details concerning each individual study are addressed in the tables Characteristics of included studies and Characteristics of excluded studies.

### Results of the search

In our original review we identified six trials that were relevant to our topic of interest (Bizzarro 2005). We included four RCTs in the review (Day 2000; Miller 2000; Morris 2000; Russell 1998) and excluded two studies (Argenziano 1998; Matsui 1997). In our first update (search to November 2007) we identified two relevant studies and excluded both (Khazin 2004; Smith 2006). For this updated search (from November 2007 to 1 December 2013) we excluded 49 studies considering the inclusion criteria and identified three additional relevant trials (Checchia 2013; Kirbas 2012; Loukhanov 2011) but none met out inclusion criteria. For a summary of the search see Figure 1.


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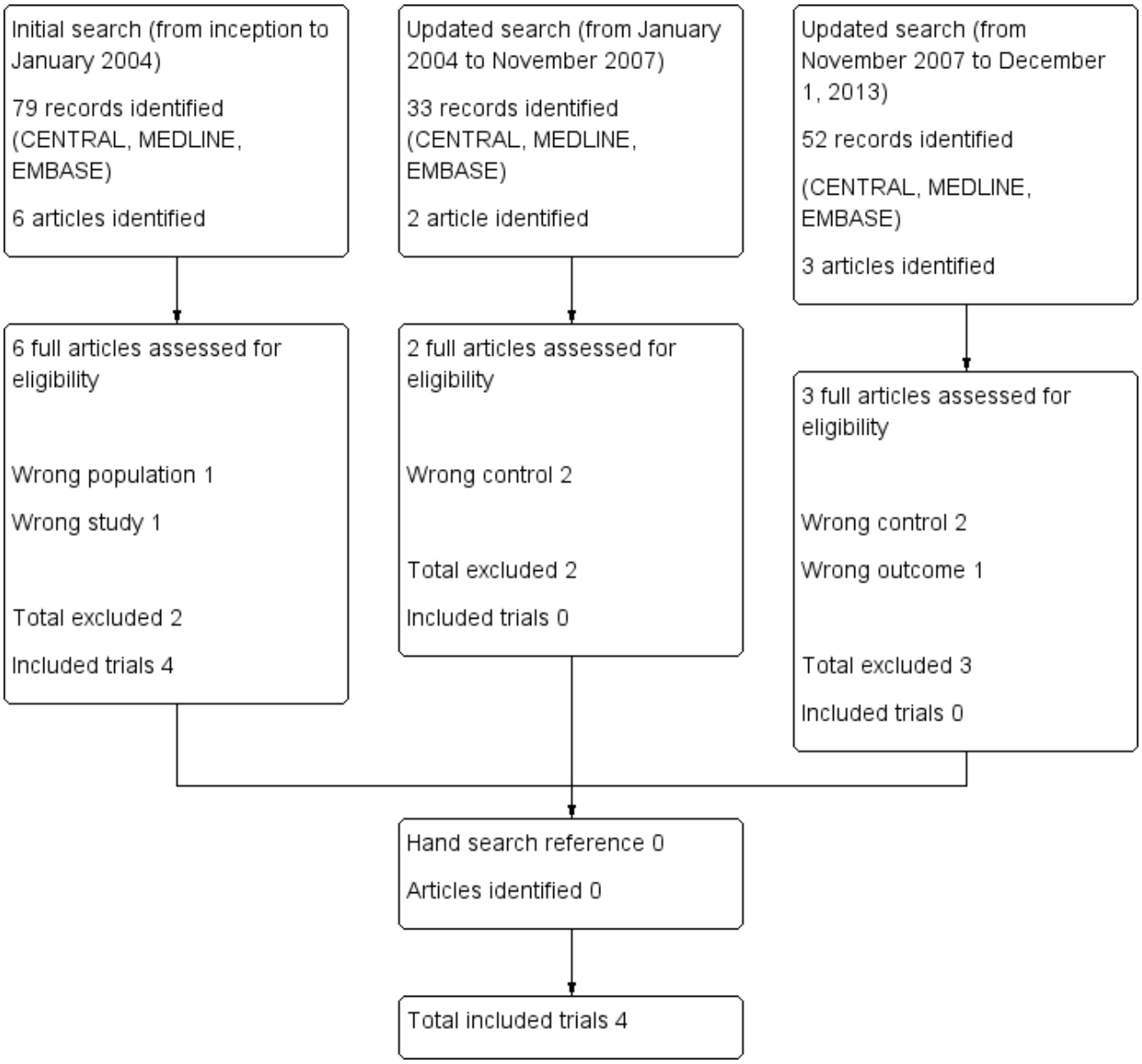
**Figure 1. Study flow diagram.**




**Included studies**

Three of the trials were of parallel design (Day 2000; Miller 2000; Russell 1998) while the fourth employed a cross-over design (Morris 2000). However, using the recommendations from The Cochrane Collaboration for evaluation of a cross-over trial (Alderson 2002), the authors concluded that the inclusion of this study into the review was valid. The details of the participants and interventions can be found in the table Characteristics of included studies.

Each of the four trials included in the review evaluated the use of postoperative iNO in infants and children with CHD necessitating repair. All included participants were identified as having pulmonary hypertension either in the preoperative or postoperative period. In three of the studies (Day 2000; Morris 2000; Russell 1998) postoperative pulmonary hypertension was used as a criterion either for inclusion or for a separate subgroup analysis. One of the RCTs (Miller 2000) utilized preoperative diagnosis as part of the entry criteria into the trial but did not comment on the presence or absence of postoperative pulmonary hypertension in individual included participants.

In each RCT children were the primary patient population studied. Russell 1998 included 35 patients aged 2 days to 6.5 years. Day 2000 included 44 patients aged 1 day to 20 years (and 2 adults). Morris 2000 included 12 patients with an age range from 1 month to 17 years (1 adult). Miller 2000 did not specifically list the age range of their patient population but instead commented on inclusion of 124 patients with a median age of 3 months in the treatment group and 2 months in the control group. We excluded data on adult patients (defined as greater than or equal to 17 years of age) from the analysis.

The RCTs differed with respect to the method of treatment utilized in the control groups. Day 2000 compared the use of iNO with 'conventional treatment' in the control group. No strict criteria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or definition of 'conventional management' were utilized in this trial and management of the control group was instead left to the discretion of the attending cardiothoracic surgeon. Morris 2000 employed a cross-over method with patients randomized to receive either iNO in the treatment group or hyperventilation as the treatment comparison in the control group (often part of the conventional management protocol). This trial utilized a 30-minute washout period between treatments. With the exception of the intervention, patient management was similar in both groups. Russell 1998 and Miller 2000 initiated iNO in the treatment group versus placebo in the control group, although underlying conventional therapies (such as hyperventilation and the use of inotropes) were maintained in both groups.

The trials also differed with respect to onset and duration of iNO therapy in the postoperative period. Russell 1998 initiated iNO or placebo therapy immediately after patients were removed from cardiopulmonary bypass and continued treatment for a total of 20 minutes. Day 2000 initiated therapy upon the patient's arrival in the intensive care unit (ICU) and continued treatment until extubation, although data collection occurred at one hour after initiation of therapy. Morris 2000 used a cross-over study design comparing iNO with hyperventilation. Treatment was initiated at a mean of 8.5 hours postoperation (range 4 to 40 hours) and each treatment was given for a period of 30 minutes. Miller 2000 initiated therapy just after completion of surgery and continued iNO or placebo for a maximum of seven days.

The concentration of iNO utilized also differed among studies. Russell 1998 initiated iNO at 80 ppm while Day 2000 initiated 20 ppm. Morris 2000 initiated a 5 ppm concentration of iNO for 15 minutes and then 40 ppm for the remaining 15 minutes of treatment (although data were presented only for outcomes on 40 ppm). Miller 2000 initiated iNO at 10 ppm and continued the use of this concentration throughout the duration of the study.

Primary outcomes differed slightly among the RCTs as well. Russell 1998 and Morris 2000 compared changes in haemodynamics (MPAP, MAP, HR) as their primary outcomes of interest. Day 2000 and Miller 2000 compared the number of PHTC and the incidence of mortality in each group as their primary outcomes of interest. Day 2000 and Miller 2000 compared changes in haemodynamic measurements as secondary outcomes of interest.

Data presentation differed among the RCTs included in this review. Russell 1998 and Morris 2000 presented individual data whereas Day 2000 and Miller 2000 presented group data. Individual data were collected from Dr Ronald Day so that certain outcomes of interest, specifically MPAP and MAP, could be calculated and included. With respect to Miller 2000, most data were presented as group data in the form of median values and interquartile ranges. We thus assumed that the data were skewed, making it difficult to calculate the necessary means and SDs required for inclusion in the analysis. Attempts to obtain the data from the authors in a form that could be utilized were unsuccessful.

### Excluded studies

We excluded seven studies during the review process. Of the excluded studies, Matsui 1997 was listed in CENTRAL but was not an RCT; Argenziano 1998 was an RCT on the topic of interest but utilized only an adult patient population; Khazin 2004 compared iNO with milrinone, not placebo or conventional management, and analysed the combined effects of the two agents on pulmonary hypertension in the patient population of interest for this review; Smith 2006 compared citrulline, an NO precursor, with placebo in the patient population of interest. Loukhanov 2011 and Kirbas 2012 compared the effects of iNO versus aerosolized iloprost on pulmonary hypertension in the patient population of interest within the first 72 hours post-bypass. Checchia 2013 compared the effects of iNO versus placebo delivered to the membrane oxygenator only during cardiopulmonary bypass in children undergoing repair of tetralogy of Fallot. The details of all exclusions can be found in Characteristics of excluded studies.

## Risk of bias in included studies

We included randomized and quasi-randomized controlled trials in this review (Characteristics of included studies). The bias domains are presented in the 'risk of bias' summary figure and in the 'risk of bias' graph (Figure 2 and Figure 3). Two studies had low risk of bias (Day 2000; Miller 2000). We assessed methodological quality based on the following criteria.


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**Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.**

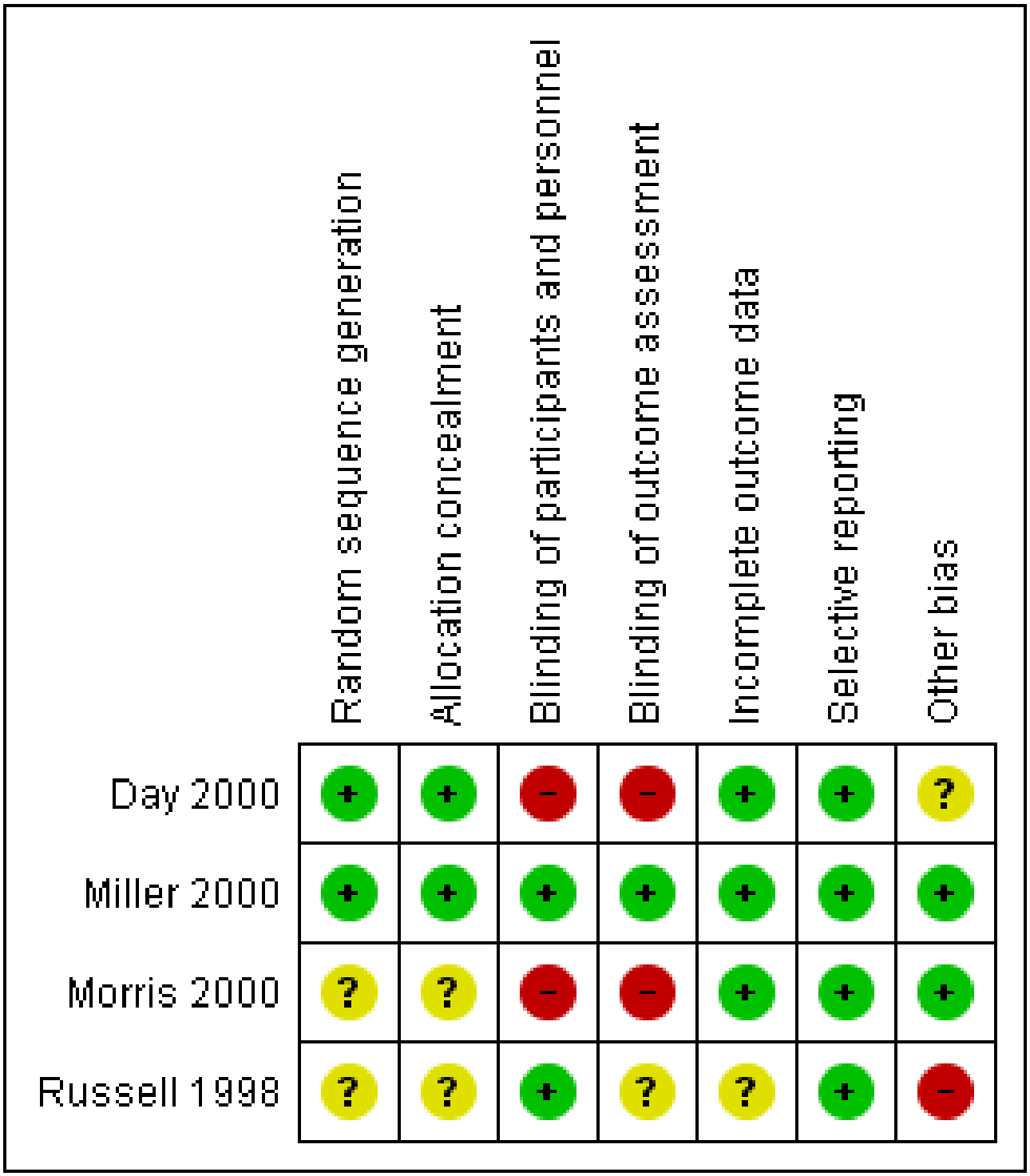



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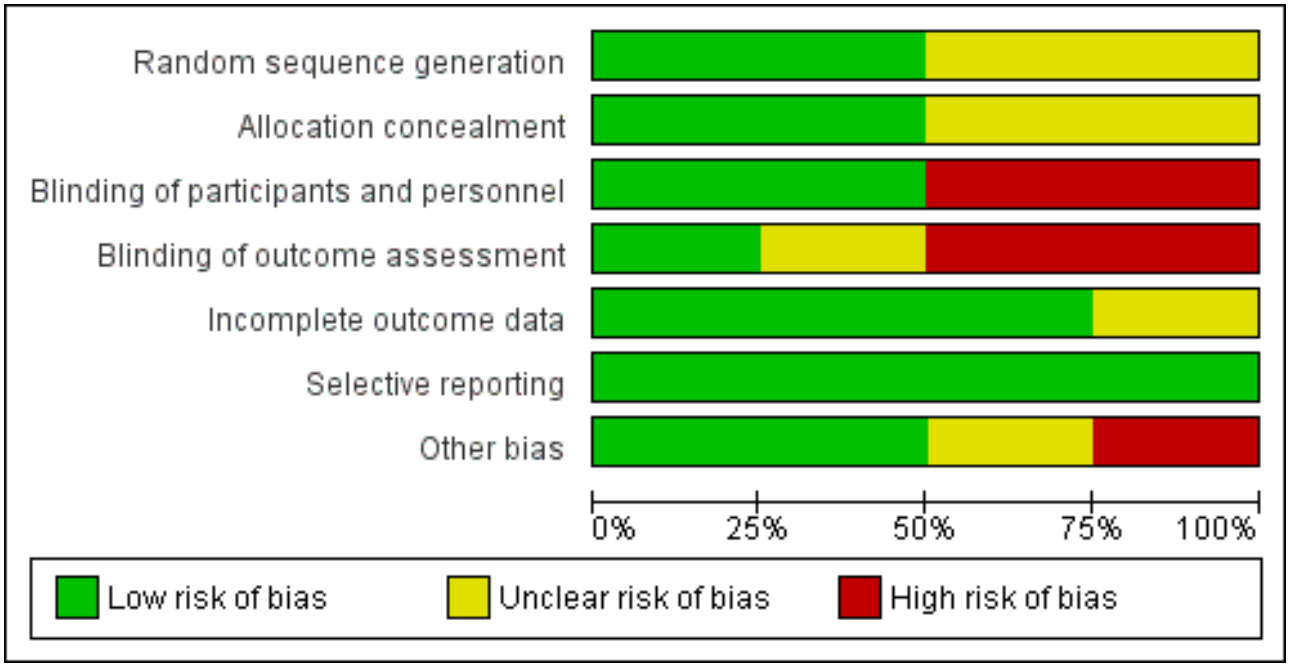
**Figure 3. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.**




**Allocation**

Russell 1998 and Morris 2000 did not list specific techniques for randomization. Day 2000 employed a "blind draw from sequential groups containing six assignments". No further detail was given as to who, the investigators or a separate participant, actually performed the draw and assignment. Finally, Miller 2000 employed a computer-based randomization program to assign patients to the treatment or control group.

Concealment of allocation was adequate in Miller 2000 and Day 2000 as blinding of randomization was likely to have been preserved through the methods of randomization. Concealment was unclear in Morris 2000 and Russell 1998 as, since no explicit details about methods of randomization were provided, the review authors had no way of confirming whether or not concealment of allocation was preserved.

**Blinding**

Russell 1998 described adequate blinding of treatment by stating, "The operative team (surgeons, anaesthesiologists, and echocardiographies) were blinded to the treatment assigned". Miller 2000 also described adequate blinding of treatment as the treatment and placebo gas "were masked with locked opaque covers" and adjusted by a non-clinical investigator. In the trials by Morris 2000 and Day 2000, however, the investigators were unblinded with respect to treatment, allowing for the possibility of performance bias. Both trials essentially employed iNO versus conventional therapy (hyperventilation in the Morris trial). Without the use of a placebo gas the investigators could not maintain blinding.

Morris 2000 and Day 2000 were unblinded with respect to outcome assessment, allowing for the introduction of detection bias into their analyses. Miller 2000 preserved blinded outcome assessment whereas Russell 1998 did not comment on this level of blinding.

We based the overall assessment of methodologic quality primarily on the quality of concealment of allocation. We also factored patient, physician, and outcome assessor blinding into the assessment. Based on this technique for evaluation, we found Miller 2000 to be of the highest methodologic quality with a low risk of bias. We found Day 2000 to be of above average quality with a low to moderate risk of bias. Although concealment of allocation was adequate, a lack of patient, physician, and outcome assessor blinding made the potential for performance and detection bias likely. We assessed Russell 1998 to be of average methodologic quality due to lack of concealment of allocation. We determined Morris 2000 to be of poor methodologic quality, particularly for the lack of concealed allocation and blinding. We attempted to assess the trials of higher quality (for example Miller 2000 and Day 2000) separately in a sensitivity analysis but could not do so due to sample size. These issues of methodologic quality are addressed in further detail in the Discussion section below.

**Incomplete outcome data**

Russell 1998 initially randomized 39 patients but only obtained results from 35 patients as four were excluded due to technical problems (three for failure to place a necessary pulmonary artery catheter and one due to equipment problems). Which group these patients were initially assigned to was not described. Day 2000 enrolled 44 patients (two patients were actually enrolled on two separate occasions for separate surgical procedures) and followed all to completion of the study. Four patients initially assigned to the control group were allowed to cross over and receive iNO if conventional therapy failed. These were excluded from the analysis. Morris 2000 enrolled 12 patients and followed all to completion. Miller 2000 initially enrolled 124 infants, all of whom completed the study. Intention to treat was preserved in all trials.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### Selective reporting

All trials reported all expected outcomes. There was no risk of bias identified in any of the studies.

### Other potential sources of bias

Two studies appeared to be free of other sources of bias (Miller 2000; Morris 2000). Day 2000 did not analyse patients with initial low pulmonary arterial pressure and excluded patients who developed an increase in pressure after they were already excluded from the study. Russell 1998 did not measure the methaemoglobin level in all patients who received iNO.

## Effects of interventions

See: **Summary of findings for the main comparison** Inhaled nitric oxide compared to placebo or conventional therapy for all infants and children with congenital heart disease having postoperative management of pulmonary hypertension

See: Summary of findings for the main comparison for the main comparison.

### Inhaled nitric oxide versus placebo or conventional treatment, or both

No data were available on several outcomes that we had hoped to include in our analysis. In particular, no long-term follow-up data were available from any of the four trials and thus we could not assess the outcomes of long-term mortality and neurodevelopmental disability. In addition, no data on length of hospital stay were available for analysis from any of the trials.

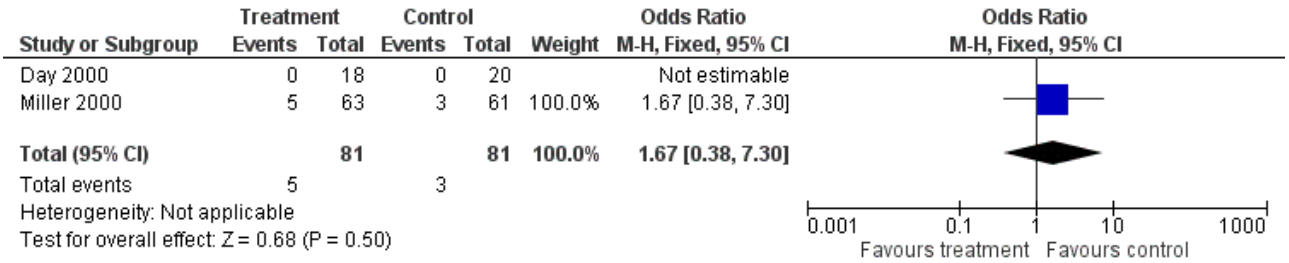
#### *Long-term mortality*

No data were available for this outcome from any of the trials included in this review.

#### *Mortality prior to discharge*

Mortality was reported in two of the four trials included in this review (Day 2000; Miller 2000). Day 2000 reported no deaths in either the treatment or control group whereas Miller 2000 observed five deaths out of the 63 children in the treatment group and three of 61 in the control group (although only one death in each group was believed to be related to a PHTC). The data presented were from parallel trials only and, therefore, we decided to calculate an OR. The analysis showed no statistically significant difference with respect to mortality between the two groups (OR 1.67, 95% CI 0.38 to 7.30; P = 0.50) (Analysis 1.1; Figure 4).

**Figure 4. Forest plot of comparison: 1 Inhaled nitric oxide versus placebo or conventional management, outcome: 1.1 Mortality prior to discharge.**




#### *Length of stay in intensive care unit or hospital*

No data were available for this outcome from any of the trials included in the review.

#### *Incidence and severity of neurodevelopmental disability*

No data were available for this outcome from any of the trials included in this review.

#### *Mean number of pulmonary hypertensive crises (PHTC)*

This outcome was reported only in Day 2000 with 3 PHTC out of the 18 children in the treatment group and 4 PHTC out of the 20 children in the control group. The data were from a parallel trial and we calculated an OR. The analysis showed no significant difference with respect to the absolute number of PHTC in the treatment versus the control groups (OR 0.80, 95% CI 0.15 to 4.18; P = 0.79) (Analysis 1.2).

#### *Difference in mean pulmonary arterial pressure (MPAP) change score*

This outcome was initially reported in two of the four trials (Morris 2000; Russell 1998) and recorded, but not reported, in a third (Day 2000). As a result, we obtained and analysed the raw, individual patient data from the investigators. A reduction in MPAP was considered a favourable outcome.

Overall, the meta-analysis revealed no significant change in MPAP between iNO and control (treatment effect -2.94 mm Hg, 95% CI -9.28 to 3.40; P = 0.36). The level of heterogeneity ($I^2$ = 45%) was just below that considered significant (Analysis 1.3).

#### *Difference in mean arterial pressure (MAP) change score*

This outcome was reported in three of the included trials (Day 2000; Morris 2000; Russell 1998). A reduction in MAP was considered an unfavourable outcome whereas no change or an increase in MAP from baseline was considered favourable.

Overall, the meta-analysis revealed no significant difference in effect between the two groups (treatment effect -3.55 mm Hg, 95% CI -11.86 to 4.76; P = 0.40). We observed no heterogeneity ($I^2$ = 0%) between trials (Analysis 1.4).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#### *Difference in heart rate (HR) change score*

This outcome was reported in three of the four trials (Day 2000; Morris 2000; Russell 1998). A reduction in HR, assuming constant stroke volume, was considered an unfavourable outcome as compared to no change or an increase from baseline.

Overall, the meta-analysis did not show a significant difference in overall effect in either group (treatment effect 0.02 bpm, 95% CI -8.13 to 8.18; P = 1.00). We observed no heterogeneity between studies ($I^2$ = 0%) (Analysis 1.5).

#### *Difference in $PaO_2$:$FiO_2$ change score*

Only one parallel trial (Day 2000) included this outcome. An increase in this ratio was considered a favourable outcome and indicative of improved arterial oxygenation. The trial observed no significant change in the ratio with the use of iNO as compared with control (MD 17.18, 95% CI -28.21 to 62.57; P = 0.46) (Analysis 1.6).

#### *Maximum methaemoglobin level*

This outcome was reported in only one of the four studies (Day 2000). An increase in methaemoglobin level was considered an unfavourable outcome. These data were from a parallel trial and thus we calculated a weighted mean difference. Overall, there was a significantly elevated methaemoglobin level in the iNO group when compared to control (MD 0.30%, 95% CI 0.24 to 0.36; P < 0.00001). However, the maximum methaemoglobin level did not approach toxicity levels (> 10%) (Analysis 1.7).

### Subgroup analyses

### Inhaled nitric oxide versus conventional management

This analysis utilized data from Day 2000 and Morris 2000. These investigators compared the use of iNO to conventional therapies in the management of postoperative pulmonary hypertension.

#### *Mortality prior to discharge*

Data from Day 2000 revealed no deaths in either treatment or control groups and so no analysis was performed.

#### *Mean number of pulmonary hypertensive crises (PHTC)*

The analysis revealed no significant difference in the number of PHTC experienced by the treatment and control groups (the same data as that presented in Analysis 1.2).

#### *Difference in MPAP change score*

This analysis included data from Day 2000 and Morris 2000. We observed no statistically significant change in MPAP (treatment effect -1.14 mm Hg, 95% CI -8.87 to 6.59; P = 0.77) and a significant amount of heterogeneity between studies ($I^2$ = 58%) (Analysis 1.3).

#### *Difference in MAP change score*

The data analysis revealed no significant difference in the change in MAP between treatment and control groups (treatment effect -1.94 mm Hg, 95% CI -11.23 to 7.35; P = 0.68). We observed no heterogeneity between studies on this outcome (Analysis 1.4).

#### *Difference in HR change score*

This analysis was again comprised of data from Day 2000 and Morris 2000. There was no significant difference observed in the change in HR between groups (treatment effect -0.37 bpm, 95% CI -8.85 to 8.10; P = 0.93). No heterogeneity was observed (Analysis 1.5).

#### *Difference in $PaO_2$:$FiO_2$ change score*

This outcome included data only from Day 2000, which was the same data as presented in Analysis 1.6 .

#### *Maximum methaemoglobin level*

This outcome was reported in only one study (Day 2000), which was the same data as presented in Analysis 1.7.

### Inhaled nitric oxide versus placebo

This analysis separated those trials utilizing placebo in the control group from those which initiated conventional therapy (that is hyperventilation, pressor support, etc). We included Miller 2000 and Russell 1998.

#### *Mortality prior to discharge*

Miller 2000 reported on this outcome. Since the only other trial (Day 2000) that also reported mortality had no events the results were the same as previously described (Analysis 1.1).

#### *Difference in MPAP change score*

We utilized data from Russell 1998 in this analysis and observed no significant change in MPAP in the treatment group compared with placebo (treatment effect -9.15 mm Hg, 95% CI -20.70 to 2.40; P = 0.12) (Analysis 1.3).

#### *Difference in MAP change score*

We utilized data from Russell 1998 in this analysis and observed no significant change in MAP with the use of iNO (treatment effect -10.00 mm Hg, 95% CI -28.60 to 8.60; P = 0.29) (Analysis 1.4).

#### *Difference in HR change score*

We utilized data from Russell 1998 in this analysis. We observed no significant difference between the treatment and control groups with respect to change in HR (treatment effect 5.00 bpm, 95% CI -24.99 to 34.99; P = 0.74) (Analysis 1.5).

### Sensitivity analysis

We considered Day 2000 and Miller 2000 to be of highest methodologic quality. Miller 2000 only reported on the outcome mortality prior to discharge. This was the same data as presented in Analysis 1.1, which showed no significant difference between groups (P = 0.50). No other sensitivity analyses could be performed.

## DISCUSSION

### Summary of main results

A substantial amount of data have been published favouring the use of iNO in the treatment of neonates with persistent pulmonary hypertension (Clark 2000; Kinsella 1992; NINOS 1997; Roberts 1992; Roberts 1997). These findings have encouraged research in other patient populations demonstrating pulmonary hypertension. In children with CHD, elevated pulmonary arterial pressures may complicate postoperative recovery and, in some instances, potentiate life-threatening crises. As a result, safe and effective therapeutic intervention has been sought. This review evaluated


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the therapeutic role of iNO in the postoperative management of infants and children with CHD.

The four RCTs included in this review focused on two major types of primary objectives. Day 2000 and Miller 2000 evaluated the postoperative effect of iNO on short-term clinical outcomes (for example mortality, number of life-threatening crises, improvements in oxygenation), while Morris 2000 and Russell 1998 evaluated the physiologic changes (for example MPAP, MAP, HR) that occur with the use of iNO. Unfortunately, no data were collected to evaluate the long-term outcomes of morbidity (for example neurodevelopmental disability) and mortality.

In analysing the primary outcome of mortality prior to discharge, we observed no statistically significant difference between the treatment and control groups. These findings are based on data from two trials (Day 2000; Miller 2000), one of which (Day 2000) observed no deaths in either group studied. While these findings indicate no substantial difference in mortality between those children treated with iNO and those treated with placebo or conventional therapy, the relatively small sample size (N = 162) and low incidence of PHTC-related postoperative deaths (N = 2) must be considered when attempting to draw general conclusions. A similar problem arises in the analysis of secondary outcomes. While no differences were observed between treatment and control groups with respect to the number of PHTC in the postoperative period, the data are based on one RCT (Day 2000).

Morris 2000 and Russell 1998 focused on changes in haemodynamics in the postoperative period. Day 2000 and Miller 2000 also documented these changes but as secondary outcomes. As a selective pulmonary vasodilator, one might expect iNO to cause a significant decrease in MPAP when compared with placebo. When we examined this outcome we observed no statistically significant change in MPAP with the use of iNO versus control. Similarly, no significant differences were observed between groups with respect to changes in MAP or HR. Subgroup analysis by method of control also demonstrated no significant differences in haemodynamics with either method of control, but valid interpretation of these findings was difficult based on the small numbers of patients analysed in each group.

The ratio $PaO_2$:$FiO_2$ is a measure of oxygenation and a positive change in this ratio implies an improvement. This outcome was measured in only one trial (Day 2000) and revealed no significant difference between treatment and control groups. It is unclear whether this lack of improvement is related to ineffectiveness of iNO to promote increased pulmonary blood flow.

The maximum methaemoglobin level, a measure of toxicity, was reported only in Day 2000. In the analysis there was a significant increase in this measure in the iNO group as compared to the control group, which one would expect as methaemoglobin, a byproduct of iNO, should not be elevated in the control group. What is more important is whether or not the maximum level approached toxicity (> 10%), which it did not.

Thus, from this analysis it appears that the postoperative use of iNO does not result in a significant reduction in mortality and the number of PHTC, nor does its use appear to significantly alter haemodynamics or result in any improvement in other clinically relevant outcomes such as arterial oxygenation. However, the validity of these conclusions needs to be examined further.

The major limitation of this review relates to sample size which, when data were available, affected both the precision and interpretability of the results. When assessing one of our primary outcomes, mortality prior to discharge, data were available from only two trials (Day 2000; Miller 2000) with only two deaths observed. Similarly, when assessing the number of PHTC, data were only available from one trial (Day 2000). Furthermore, data on long-term mortality, neurodevelopmental disability, and length of stay were simply not available. In addition, there was significant potential for selection bias in two of the four trials (Morris 2000; Russell 1998) and heterogeneity between studies given that all trials utilized different concentrations of inhaled nitric oxide, different lengths of administration of therapy, and included patients with diverse congenital heart lesions and of various ages.

The strengths of the review were that it included four RCTs. Also, we were able to obtain data on the majority of our outcomes of interest, which included individual patient data in most cases. Lastly, while there was concern about heterogeneity, no significant heterogeneity was observed between studies when analysing the majority of outcomes of interest.

### Overall completeness and applicability of evidence

Several factors exist within and amongst the included trials that make an accurate meta-analysis with a definitive conclusion difficult. The first area of concern to be addressed is with respect to the methodologic quality of the included studies. The trial of highest methodologic quality (Miller 2000) only included data with respect to mortality that could be included in the analysis. Of the three remaining trials, we assessed only one (Day 2000) as being of relatively high methodologic quality based on concealment of allocation. Thus, the potential for selection bias is likely in the remaining two trials (Morris 2000; Russell 1998). Although the results of this meta-analysis do not appear to show any significant clinical benefit of iNO to treat pulmonary hypertension in children with CHD, any significant clinical harm is not identified in this systematic review.

### Quality of the evidence

We considered the quality of the evidence using the measurement of risk of bias. There was one study that had a low risk of bias in all domains (Miller 2000). Random sequence generation, allocation concealment, incomplete outcome data, and selective reporting were not assessed as high risk of bias in all included studies. Small sample sizes affected precision of the results. There was not substantial heterogeneity between the studies that analysed the majority of outcomes of interest.

### Potential biases in the review process

Factors that also contribute to potential biases are lack of blinding to the intervention (Day 2000; Morris 2000) and lack of blinded outcome assessment, or failure to report this procedure, in all three studies. Therefore, the potential for the introduction of bias into the trials and subsequently the analysis is likely. We attempted to address this issue via sensitivity analysis, by evaluating only those trials of high methodologic quality (Day 2000; Miller 2000). Unfortunately the only outcome that could be assessed, given the lack of individual data from one trial (Miller 2000), was mortality prior to discharge.


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Also of concern is the relatively sparse number of patients included in the majority of the studies, as documented in the 'Additional tables' section of this review (see Table 1; Table 2; Table 3). A total of 73 outcomes collected from 62 infants and children (iNO group = 34, placebo and conventional management groups = 39) were included (the discrepancy arises as a result of the cross-over design of Morris 2000). A possible explanation for the relatively small numbers of participants lies in the fact that we are dealing with a very specific population with CHD necessitating repair and demonstrating elevated pulmonary arterial pressures. The small sample sizes are reflected in the wide CIs seen in the analysis and, as a result, compromise the precision of the review.

Another serious problem with the meta-analysis concerns the potential heterogeneity that exists among the included investigations. All four trials utilized different concentrations of iNO, different lengths of administration of therapy, and included patients with diverse congenital heart lesions and of various ages. While the $I^2$ statistic for heterogeneity is not statistically significant in most cases, the potential is apparent. An attempt to deal with this is presented in the form of the subgroup analysis on method of control. Unfortunately, further subgroup analyses based on host factors and concentration of iNO administered could not be performed due to small sample sizes and missing individual patient data. Therefore, although no significant differences were observed with the use of iNO in this patient population, there is the potential for heterogeneity and bias that compromise the results of the review and make it impossible to draw any definitive conclusions at the present time.

### Agreements and disagreements with other studies or reviews

A Cochrane systematic review considered inhaled nitric oxide for respiratory failure in infants born at or near term (Finer 2009). The authors found a statistically significant reduction in death or need for extracorporeal membrane oxygenation (ECMO) with the use of iNO (Mantel-Hanzel risk ratio 0.65, 95% CI 0.55 to 0.76). This paper did not include studies from our review as it focused on infants with acute respiratory disease, not postoperative patients with CHD, and assessed different outcomes of interest including the need for ECMO.

## AUTHORS' CONCLUSIONS

### Implications for practice

The results of this meta-analysis do not appear to show any significant clinical benefit with the use of postoperative iNO to treat pulmonary hypertension in children with CHD. We have observed no differences with respect to mortality, number of PHTC, arterial oxygenation, or changes in haemodynamics with the use of postoperative iNO. There are no data to determine the effects of treatment with iNO on length of stay in an ICU or hospital, long-term mortality, or neurodevelopmental disability. Furthermore, the relatively sparse patient population and the potential biases that may be present as a result of problems with the methodology of the trials make it impossible to draw any valid conclusions about treatment effect at this time.

### Implications for research

It is evident from this review that more data from randomized controlled trials of high methodologic quality are necessary in order to appropriately evaluate the therapeutic potential of iNO in this clinical setting. Also, although the outcomes reviewed in these trials are of clinical importance, more relevant outcomes need to be assessed. In particular, additional patient data with respect to mortality; length of ICU and hospital stay, or both; neurodevelopmental outcomes; and measures of oxygenation necessitate further investigation.

## ACKNOWLEDGEMENTS

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We would like to thank Dr Ronald Day for the contribution of his individual patient data for analysis and inclusion in this review.

We would also like to thank Dr Michael Bracken for his instruction in the concepts and principles of evidence-based medicine and the preparation of a systematic review in addition to his statistical assistance with the analyses of data.


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* Indicates the major publication for the study

# CHARACTERISTICS OF STUDIES

## Characteristics of included studies *[ordered by study ID]*

### Day 2000

| | |
|---|---|
| Methods | Parallel randomized controlled trial |
| Participants | • 38 participants randomized<br>• Included: patients with congenital heart disease who underwent a biventricular repair or heart transplantation with systolic pulmonary arterial pressure 50% higher than systolic systemic arterial pressure<br>• Excluded: not reported<br>• Age: months (median 6 with range from 1 day to 3 years in control group; 7 with range from 1 day to 20 years in intervention group)<br>• Sex distribution: not reported<br>• Country: United States of America |
| Interventions | 1. iNO at 20 ppm. Therapy was initiated in the ICU and continued until extubation. Data were collected at 1 hour after initiation of therapy<br>2. Conventional management as dictated by the attending cardiothoracic surgeon |
| Outcomes | Primary: Number of pulmonary hypertensive crises<br>Secondary: Changes in pH, $PCO_2$, $PaO_2/FiO_2$, heart rate, systemic and pulmonary artery pressures, in addition to the maximum methaemoglobin concentration |
| Notes | Single centre<br>Several patients had radiographic evidence of lung disease before operation in each group<br>Could be included in the following subgroup analyses: conventional management |

***Risk of bias***

| Bias | Authors' judgement | Support for judgement |
|---|---|---|
| Random sequence generation | Low risk | Quote: "Patients were randomized by a blind draw from sequential groups containing six assignments."<br>Comment: Probably done |
| Allocation concealment | Low risk | Quote: "Patients were randomized by a blind draw from sequential groups containing six assignments."<br>Comment: Probably done |
| Blinding of participants and personnel<br>All outcomes | High risk | Quote: "Inhaled nitric oxide was then administered to patients in the treatment group in an open labelled, or unblinded manner as previously described ..."<br>Comment: Not done |
| Blinding of outcome assessment<br>All outcomes | High risk | Quote: "Inhaled nitric oxide was then administered to patients in the treatment group in an open labelled, or unblinded manner as previously described ..."<br>Comment: Not done |


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**Day 2000** *(Continued)*

| | | |
|---|---|---|
| Incomplete outcome data<br>All outcomes | Low risk | 6/44 missing data. 2 were enrolled twice. Two patients were excluded for age > 14 years and 4 were excluded due to cross over. In the final analysis, there were a total of n = 18 treatment and n = 20 controls |
| Selective reporting | Low risk | All expected outcomes were reported |
| Other bias | Unclear risk | Quote: "We may have failed to identify all patients with a systolic pulmonary arterial pressure of 50% or more of the systolic systemic pressure, and some patients with an initially low pulmonary arterial pressure may have developed a subsequent increase in pressure that was not monitored."<br>Comment: Eligible patients could not be included in this study |

**Miller 2000**

| | |
|---|---|
| Methods | Parallel randomized controlled trial |
| Participants | • 124 participants randomized<br>• Included: infants suitable for corrective heart surgery with high pulmonary flow, pressure, or both, congenital heart lesions, such as non-restrictive ventricular septal defect, complete atrioventricular septal defect, truncus arteriosus, or total anomalous pulmonary venous drainage, with objective evidence of pulmonary hypertension at the immediate preoperative assessment<br>• Excluded: not reported<br>• Age: months (median 3 with range from 1 to 5 years in treatment group; 2 with range from 1 to 4 in control group)<br>• Sex distribution: M/F (36/27, 28/33)<br>• Country: Australia |
| Interventions | 1. iNO at 10 ppm<br>2. Placebo management. Nitrogen was used |
| Outcomes | Primary: Number of pulmonary hypertensive crises and deaths<br>Secondary: Changes in pulmonary and systemic artery pressures as well as pulmonary and systemic vascular resistance indexes, duration of mechanical ventilation, and maximum methaemoglobin level |
| Notes | Single centre<br>Could be included in the following subgroup analyses: placebo management |

***Risk of bias***

| Bias | Authors' judgement | Support for judgement |
|---|---|---|
| Random sequence generation | Low risk | Quote: "Infants were randomised locally by a computer-based stratified minimisation algorithm, based on the presence or absence of Down's syndrome ..."<br>Comment: Clear. Authors employed a computer-based randomization program |
| Allocation concealment | Low risk | Quote: "On the two types of nitric oxide and nitrogen dioxide monitors, the displays were masked with locked opaque covers."<br>Comment: Clear |
| Blinding of participants and personnel<br>All outcomes | Low risk | Quote: "A non-clinical investigator who was aware of treatment status was available around the clock to supervise randomisation, preparation of the cylinders of study gas, and to set up gas delivery and the monitoring circuit for each patient." |


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**Miller 2000** *(Continued)*

| | | |
|---|---|---|
| | | Comment: Clear |
| Blinding of outcome assessment<br>All outcomes | Low risk | Quote: "The latter radiographs were reviewed independently by a chest physician and radiologist who were unaware of study-gas assignment, and given a predefined lung injury score ..."<br>Quote: "When infants were eligible for extubation, the nonclinical investigator reduced the study-gas flow by 20% per h, aiming for complete withdrawal after 4 h."<br>Comment: Probably done |
| Incomplete outcome data<br>All outcomes | Low risk | No missing data. All patients analysed |
| Selective reporting | Low risk | All expected outcomes were reported |
| Other bias | Low risk | The study appears to be free of other sources of bias |

**Morris 2000**

| | |
|---|---|
| Methods | Cross-over randomized controlled trial |
| Participants | • 12 participants randomized<br>• Included: patients with age 1 month to 17 years, with CHD necessitating repair and with pulmonary hypertension in the postoperative period<br>• Excluded: not reported<br>• Age: years (authors reported age of each patient)<br>• Sex distribution: not reported<br>• Country: Canada |
| Interventions | 1. iNO at 5 ppm for 15 minutes followed by 40 ppm for 15 minutes. Therapy was initiated upon arrival in the ICU and continued for 30 minutes<br>2. Placebo management. Hyperventilation in the control group |
| Outcomes | Primary: Changes in mean pulmonary artery pressure and pulmonary vascular resistance index<br>Secondary: Changes in heart rate, cardiac index, stroke index, central venous pressure, left atrial pressure, and mean blood pressure |
| Notes | Single centre<br>Could be included in the following subgroup analyses: conventional management |

***Risk of bias***

| **Bias** | **Authors' judgement** | **Support for judgement** |
|---|---|---|
| Random sequence generation | Unclear risk | Quote: "The design was a crossover study with the order of the treatments randomized."<br>Comment: Authors did not list specific techniques for randomization |
| Allocation concealment | Unclear risk | Comment: Not clear |
| Blinding of participants and personnel<br>All outcomes | High risk | Comment: Probably not done |


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**Morris 2000** *(Continued)*

| | | |
|---|---|---|
| Blinding of outcome assessment<br>All outcomes | High risk | Comment: Probably not done |
| Incomplete outcome data<br>All outcomes | Low risk | Comment: 1 patient was excluded due to age > 14 years. A cross-over design was employed so that 11 patients were subsequently analysed in each group. No missing data |
| Selective reporting | Low risk | Comment: All expected outcomes were reported |
| Other bias | Low risk | The study appears to be free of other sources of bias |

**Russell 1998**

| | |
|---|---|
| Methods | Parallel randomized controlled trial |
| Participants | • 35 patients randomized<br>• Included: patients with preoperative pulmonary hypertension diagnosed by either cardiac catheterization or echocardiography<br>• Excluded: not reported<br>• Age: not reported<br>• Sex distribution: not reported<br>• Country: United States of America |
| Interventions | 1. iNOat 80 ppm. Therapy was initiated in the operating room and continued for 20 minutes<br>2. Placebo management |
| Outcomes | Primary: Changes in mean pulmonary artery pressure. Secondary: Changes in heart rate and mean arterial pressure. Median methaemoglobin level was also assessed |
| Notes | Single centre<br>Could be included in the following subgroup analyses: placebo management |

***Risk of bias***

| **Bias** | **Authors' judgement** | **Support for judgement** |
|---|---|---|
| Random sequence generation | Unclear risk | Quote: "Patients were randomly assigned to receive nitrogen (placebo) or inhaled NO (diluted in a carrier gas of nitrogen) at an inspired oxygen concentration of 0.90."<br><br>Comment: Authors did not list specific techniques for randomization |
| Allocation concealment | Unclear risk | Comment: Not clear |
| Blinding of participants and personnel<br>All outcomes | Low risk | Quote: "The operative team (surgeons, anaesthesiologists, and echocardiographers) were blinded to the treatment assigned."<br>Comment: Clear |
| Blinding of outcome assessment<br>All outcomes | Unclear risk | Quote: “The operative team (surgeons, anaesthesiologists, and echocardiographies) were blinded to the treatment assigned)”<br>Comment: Authors did not comment on this level of blinding |
| Incomplete outcome data<br>All outcomes | Unclear risk | 4/39 missing data |


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**Russell 1998** *(Continued)*

| | | |
|---|---|---|
| | | Quote: "Results were obtained in 36 of 40 studies; 4 studies were excluded from the final analysis because of technical difficulties with equipment (n = 1) or surgical decision not to use a pulmonary artery catheter (n = 3). The final data set was thus comprised of 36 studies from 35 patients ... "<br>Comment: Which group these patients were initially assigned to was not described. One patient was analysed twice (at different times) |
| Selective reporting | Low risk | Comment: All expected outcomes were reported |
| Other bias | High risk | Quote: "Methemoglobin levels were measured in 9 patients who received inhaled NO and 11 patients who did not."<br>Comment: Methemoglobin analysis was compromised |

Congenital heart disease (CHD)
Female (F)
Inhaled nitric oxide (iNO)
Intensive care unit (ICU)
Male (M)
Nitric oxide (NO)
Number of patients (n)
Parts per million (ppm)

**Characteristics of excluded studies** *[ordered by study ID]*

| Study | Reason for exclusion |
|---|---|
| Argenziano 1998 | The trial utilized an adult patient population |
| Checchia 2013 | The trial compared inhaled nitric oxide with placebo during cardiopulmonary bypass, not in the postoperative period, and did not assess pulmonary hypertension as an outcome |
| Khazin 2004 | The trial compared nitric oxide with milrinone and in combination with milrinone, not in placebo or conventional management |
| Kirbas 2012 | The trial compared inhaled nitric oxide with aerosolized iloprost, not in placebo or conventional management |
| Loukhanov 2011 | The trial compared inhaled nitric oxide with aerosolized iloprost, not in placebo or conventional management |
| Matsui 1997 | The study was not a randomized clinical trial |
| Smith 2006 | The trial compared citrulline, a nitric oxide precursor, with placebo |

## DATA AND ANALYSES


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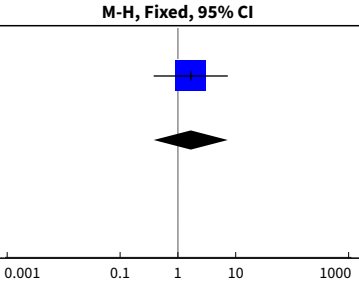
**Comparison 1. Inhaled nitric oxide versus placebo or conventional management**

| Outcome or subgroup title | No. of studies | No. of partici-pants | Statistical method | Effect size |
|---|---|---|---|---|
| 1 Mortality prior to discharge | 2 | 162 | Odds Ratio (M-H, Fixed, 95% CI) | 1.67 [0.38, 7.30] |
| 2 Pulmonary hypertensive crises | 1 | 38 | Odds Ratio (M-H, Fixed, 95% CI) | 0.8 [0.15, 4.18] |
| 3 Difference in MPAP change score (SE) | 3 | | Change in MPAP (Random, 95% CI) | -2.94 [-9.28, 3.40] |
| 3.1 Conventional Management | 2 | | Change in MPAP (Random, 95% CI) | -1.14 [-8.87, 6.59] |
| 3.2 Placebo | 1 | | Change in MPAP (Random, 95% CI) | -9.15 [-20.70, 2.40] |
| 4 Difference in MAP change score (SE) | 3 | | Change in MAP (Fixed, 95% CI) | -3.55 [-11.86, 4.76] |
| 4.1 Conventional Mangement | 2 | | Change in MAP (Fixed, 95% CI) | -1.94 [-11.23, 7.35] |
| 4.2 Placebo | 1 | | Change in MAP (Fixed, 95% CI) | -10.0 [-28.60, 8.60] |
| 5 Difference in HR change score (SE) | 3 | | Change in HR (Fixed, 95% CI) | 0.02 [-8.13, 8.18] |
| 5.1 Conventional Management | 2 | | Change in HR (Fixed, 95% CI) | -0.37 [-8.85, 8.10] |
| 5.2 Placebo | 1 | | Change in HR (Fixed, 95% CI) | 5.0 [-24.99, 34.99] |
| 6 Difference in PaO2:FiO2 change score | 1 | 38 | Mean Difference (IV, Fixed, 95% CI) | 17.18 [-28.21, 62.57] |
| 7 Maximum methaemoglobin level | 1 | 38 | Mean Difference (IV, Fixed, 95% CI) | 0.30 [0.24, 0.36] |

**Analysis 1.1. Comparison 1 Inhaled nitric oxide versus placebo or conventional management, Outcome 1 Mortality prior to discharge.**

| Study or subgroup | Treatment n/N | Control n/N | Odds Ratio M-H, Fixed, 95% CI | Weight | Odds Ratio M-H, Fixed, 95% CI |
|---|---|---|---|---|---|
| Day 2000 | 0/18 | 0/20 | | | Not estimable |
| Miller 2000 | 5/63 | 3/61 | | 100% | 1.67[0.38,7.3] |
| **Total (95% CI)** | **81** | **81** | | **100%** | **1.67[0.38,7.3]** |
| Total events: 5 (Treatment), 3 (Control) | | | | | |
| Heterogeneity: Not applicable | | | | | |
| Test for overall effect: Z=0.68(P=0.5) | | | | | |





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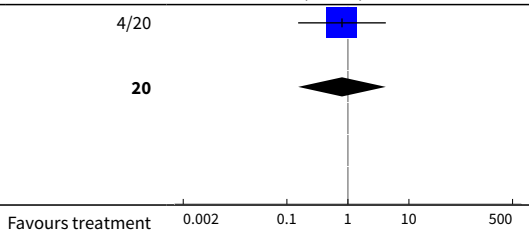
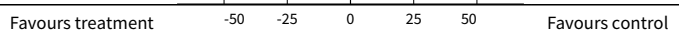
## Analysis 1.2. Comparison 1 Inhaled nitric oxide versus placebo or conventional management, Outcome 2 Pulmonary hypertensive crises.

| Study or subgroup | Treatment n/N | Control n/N | Odds Ratio M-H, Fixed, 95% CI | Weight | Odds Ratio M-H, Fixed, 95% CI |
|---|---|---|---|---|---|
| Day 2000 | 3/18 | 4/20 | | 100% | 0.8[0.15,4.18] |
| **Total (95% CI)** | **18** | **20** | | **100%** | **0.8[0.15,4.18]** |
| Total events: 3 (Treatment), 4 (Control) | | | | | |
| Heterogeneity: Not applicable | | | | | |
| Test for overall effect: Z=0.26(P=0.79) | | | | | |



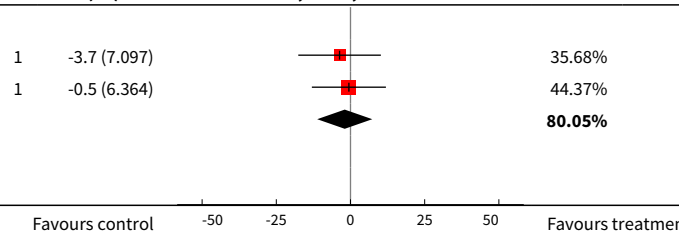

## Analysis 1.3. Comparison 1 Inhaled nitric oxide versus placebo or conventional management, Outcome 3 Difference in MPAP change score (SE).

| Study or subgroup | Treatment N | Control N | Change in MPAP (SE) | Change in MPAP IV, Random, 95% CI | Weight | Change in MPAP IV, Random, 95% CI |
|---|---|---|---|---|---|---|
| **1.3.1 Conventional Management** | | | | | | |
| Day 2000 | 1 | 1 | -4.5 (2.82) | | 47.12% | -4.46[-9.99,1.07] |
| Morris 2000 | 1 | 1 | 3.6 (4.353) | | 31.52% | 3.55[-4.98,12.08] |
| **Subtotal (95% CI)** | | | | | **78.63%** | **-1.14[-8.87,6.59]** |
| Heterogeneity: Tau²=18.63; Chi²=2.38, df=1(P=0.12); I²=58.06% | | | | | | |
| Test for overall effect: Z=0.29(P=0.77) | | | | | | |
| **1.3.2 Placebo** | | | | | | |
| Russell 1998 | 1 | 1 | -9.1 (5.893) | | 21.37% | -9.15[-20.7,2.4] |
| **Subtotal (95% CI)** | | | | | **21.37%** | **-9.15[-20.7,2.4]** |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z=1.55(P=0.12) | | | | | | |
| **Total (95% CI)** | | | | | **100%** | **-2.94[-9.28,3.4]** |
| Heterogeneity: Tau²=14.26; Chi²=3.62, df=2(P=0.16); I²=44.75% | | | | | | |
| Test for overall effect: Z=0.91(P=0.36) | | | | | | |
| Test for subgroup differences: Chi²=1.28, df=1 (P=0.26), I²=21.58% | | | | | | |




## Analysis 1.4. Comparison 1 Inhaled nitric oxide versus placebo or conventional management, Outcome 4 Difference in MAP change score (SE).

| Study or subgroup | Treatment N | Control N | Change in MAP (SE) | Change in MAP IV, Fixed, 95% CI | Weight | Change in MAP IV, Fixed, 95% CI |
|---|---|---|---|---|---|---|
| **1.4.1 Conventional Mangement** | | | | | | |
| Day 2000 | 1 | 1 | -3.7 (7.097) | | 35.68% | -3.68[-17.59,10.23] |
| Morris 2000 | 1 | 1 | -0.5 (6.364) | | 44.37% | -0.54[-13.01,11.93] |
| **Subtotal (95% CI)** | | | | | **80.05%** | **-1.94[-11.23,7.35]** |
| Heterogeneity: Tau²=0; Chi²=0.11, df=1(P=0.74); I²=0% | | | | | | |
| Test for overall effect: Z=0.41(P=0.68) | | | | | | |





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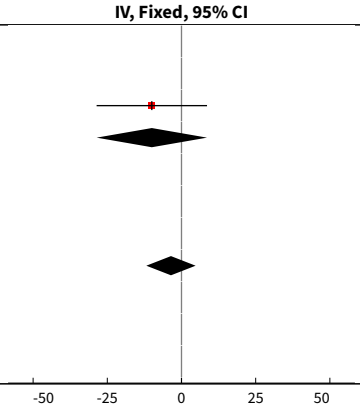
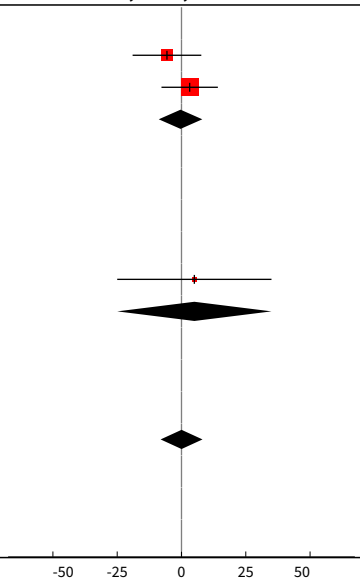
| Study or subgroup | Treatment N | Control N | Change in MAP (SE) | Change in MAP IV, Fixed, 95% CI | Weight | Change in MAP IV, Fixed, 95% CI |
|---|---|---|---|---|---|---|
| **1.4.2 Placebo** | | | | | | |
| Russell 1998 | 1 | 1 | -10 (9.491) | | 19.95% | -10[-28.6,8.6] |
| **Subtotal (95% CI)** | | | | | **19.95%** | **-10[-28.6,8.6]** |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z=1.05(P=0.29) | | | | | | |
| **Total (95% CI)** | | | | | **100%** | **-3.55[-11.86,4.76]** |
| Heterogeneity: Tau²=0; Chi²=0.69, df=2(P=0.71); I²=0% | | | | | | |
| Test for overall effect: Z=0.84(P=0.4) | | | | | | |
| Test for subgroup differences: Chi²=0.58, df=1 (P=0.45), I²=0% | | | | | | |



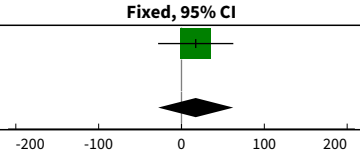

## Analysis 1.5. Comparison 1 Inhaled nitric oxide versus placebo or conventional management, Outcome 5 Difference in HR change score (SE).

| Study or subgroup | Treatment N | Control N | Change in HR (SE) | Change in HR IV, Fixed, 95% CI | Weight | Change in HR IV, Fixed, 95% CI |
|---|---|---|---|---|---|---|
| **1.5.1 Conventional Management** | | | | | | |
| Day 2000 | 1 | 1 | -5.6 (6.815) | | 37.26% | -5.65[-19.01,7.71] |
| Morris 2000 | 1 | 1 | 3.2 (5.592) | | 55.34% | 3.18[-7.78,14.14] |
| **Subtotal (95% CI)** | | | | | **92.61%** | **-0.37[-8.85,8.1]** |
| Heterogeneity: Tau²=0; Chi²=1, df=1(P=0.32); I²=0.32% | | | | | | |
| Test for overall effect: Z=0.09(P=0.93) | | | | | | |
| **1.5.2 Placebo** | | | | | | |
| Russell 1998 | 1 | 1 | 5 (15.302) | | 7.39% | 5[-24.99,34.99] |
| **Subtotal (95% CI)** | | | | | **7.39%** | **5[-24.99,34.99]** |
| Heterogeneity: Not applicable | | | | | | |
| Test for overall effect: Z=0.33(P=0.74) | | | | | | |
| **Total (95% CI)** | | | | | **100%** | **0.02[-8.13,8.18]** |
| Heterogeneity: Tau²=0; Chi²=1.12, df=2(P=0.57); I²=0% | | | | | | |
| Test for overall effect: Z=0.01(P=1) | | | | | | |
| Test for subgroup differences: Chi²=0.11, df=1 (P=0.74), I²=0% | | | | | | |




## Analysis 1.6. Comparison 1 Inhaled nitric oxide versus placebo or conventional management, Outcome 6 Difference in PaO2:FiO2 change score.

| Study or subgroup | Treatment N | Treatment Mean(SD) | Control N | Control Mean(SD) | Mean Difference Fixed, 95% CI | Weight | Mean Difference Fixed, 95% CI |
|---|---|---|---|---|---|---|---|
| Day 2000 | 18 | 39 (53.4) | 20 | 21.9 (86.9) | | 100% | 17.18[-28.21,62.57] |
| **Total ***** | **18** | | **20** | | | **100%** | **17.18[-28.21,62.57]** |





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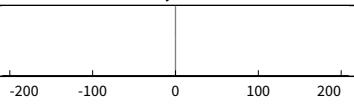
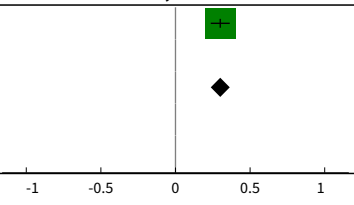
| Study or subgroup | Treatment N | Treatment Mean(SD) | Control N | Control Mean(SD) | Mean Difference Fixed, 95% CI | Weight | Mean Difference Fixed, 95% CI |
|---|---|---|---|---|---|---|---|
| Heterogeneity: Not applicable | | | | | | | |
| Test for overall effect: Z=0.74(P=0.46) | | | | | | | |




**Analysis 1.7. Comparison 1 Inhaled nitric oxide versus placebo or conventional management, Outcome 7 Maximum methaemoglobin level.**

| Study or subgroup | Treatment N | Treatment Mean(SD) | Control N | Control Mean(SD) | Mean Difference Fixed, 95% CI | Weight | Mean Difference Fixed, 95% CI |
|---|---|---|---|---|---|---|---|
| Day 2000 | 18 | 1.4 (0.1) | 20 | 1.1 (0.1) | | 100% | 0.3[0.24,0.36] |
| **Total ***** | **18** | | **20** | | | **100%** | **0.3[0.24,0.36]** |
| Heterogeneity: Not applicable | | | | | | | |
| Test for overall effect: Z=9.23(P<0.0001) | | | | | | | |




## ADDITIONAL TABLES

**Table 1. Individual number of entries for outcome, "Difference in MPAP change score (SE)"**

| Trial | iNO (N) | Control (N) |
|---|---|---|
| Russell 1998 | 5 | 8 |
| Day 2000 | 17 | 20 |
| Morris 2000 | 11 | 11 |

Number of entries (N)

**Table 2. Individual number of entries for outcome,"Difference in MAP change score (SE)"**

| Trial | iNO (N) | Control (N) |
|---|---|---|
| Russell 1998 | 5 | 8 |
| Day 2000 | 18 | 20 |
| Morris 2000 | 11 | 11 |

Number of entries (N)

**Table 3. Individual number of entries for outcome, 'Difference in HR change score (SE)'**

| Trial | iNO (N) | Control (N) |
|---|---|---|


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**Table 3. Individual number of entries for outcome, 'Difference in HR change score (SE)'** *(Continued)*

| | | |
|---|---|---|
| Russell 1998 | 5 | 8 |
| Day 2000 | 12 | 17 |
| Morris 2000 | 11 | 11 |

Number of entries (N)

## APPENDICES

### Appendix 1. Search strategy for CENTRAL

| |
|---|
| #1 HEART DEFECTS CONGENITAL |
| #2 (congenital near heart) |
| #3 (heart near defect*) |
| #4 (heart near abnormal*) |
| #5 (#1 or #2 or #3 or #4) |
| #6 NITRIC OXIDE |
| #7 (nitric next oxide) |
| #8 ino |
| #9 (#6 or #7 or #8) |
| #10 (#5 and #9) |

### Appendix 2. Search strategy for MEDLINE

| |
|---|
| #1 exp Heart Defects, Congenital/ |
| #2 (congenital$ adj3 heart).tw. |
| #3 (heart adj3 defect$).tw. |
| #4 (heart adj3 abnormal$).tw. |
| #5 or/1-4 |
| #6 Nitric oxide/ |
| #7 nitric oxide.tw. |
| #8 ino.tw. |
| #9 or/6-8 |


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*(Continued)*

| |
|---|
| #10 5 and 9 |
| + RCT filter terms. |

## Appendix 3. Search strategy for EMBASE

| |
|---|
| #1 exp CHD/ |
| #2 (congenital$ adj3 heart).tw. |
| #3 (heart adj3 defect$).tw. |
| #4 (heart adj3 abnormal$).tw. |
| #5 or/1-4 |
| #6 Nitric oxide/ |
| #7 nitric oxide.tw. |
| #8 ino.tw. |
| #9 or/6-8 |
| #10 5 and 9 |
| #11 random$.ti,ab. |
| #12 factorial$.ti,ab. |
| #13 (crossover$ or cross over$ or cross-over$).ti,ab. |
| #14 placebo$.ti,ab. |
| #15 (double$ adj blind$).ti,ab. |
| #16 (singl$ adj blind$).ti,ab. |
| #17 assign$.ti,ab. |
| #18 allocat$.ti,ab. |
| #19 volunteer$.ti,ab. |
| #20 Crossover Procedure/ |
| #21 Double Blind Procedure/ |
| #22 Randomized Controlled Trial/ |
| #23 Single Blind Procedure/ |


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*(Continued)*

| |
|---|
| #24 or/11-23 |
| #25 exp animal/ |
| #26 nonhuman/ |
| #27 exp animal experiment/ |
| #28 or/25-27 |
| #29 exp human/ |
| #30 28 not 29 |
| #31 24 not 30 |
| #32 10 and 31 |

## Appendix 4. Data extraction form

DATA EXTRACTION FORM

REVIEW ID

| |
|---|
| Study Id: |
| Authors: |
| Medline Journal Id: |
| Year of Publication: |
| Language: |
| Type of study:<br>RCT ( )<br>CCT ( )<br>Non-randomized ( ) |
| Comments on Study Design: |

Quality of Concealment of Allocation

| Quality of Concealment of Allocation | Points |
|---|---|
| Allocation was not concealed (e.g. quasi-randomization) | 0 |


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*(Continued)*

| | |
|---|---|
| Allocation concealment was not stated or was unclear | 1 |
| Disclosure of allocation was a possibility | 2 |
| Allocation was concealed (e.g. numbered, sealed opaque envelopes drawn NON consecutively) | 3 |
| | |
| Inclusion and exclusion criteria were not clearly defined in the text | 0 |
| Inclusion and exclusion criteria were clearly defined in the text | 1 |
| | |
| Outcomes of patients who withdrew or were excluded after allocation were NEITHER detailed separately NOR included in an intention to treat | 0 |
| Outcomes of patients who withdrew or were excluded after allocation were EITHER detailed separately OR included in an intention to treat analysis OR the text stated there were no withdrawals | 1 |
| | |
| Treatment and control groups were NOT adequately described at entry | 0 |
| Treatment and control groups were adequately described at entry<br>A minimum of 7 admission details were described<br>(e.g. age, sex, mobility, type of surgery, ASA grade, function score, mental test score | 1 |
| | |
| The text stated that the care programmes other than trial options were NOT identical | 0 |
| The text stated that the care programmes other than trial options were identical | 1 |
| | |
| Outcome measures were NOT clearly defined in the text | 0 |
| Outcome measures were clearly defined in the text | 1 |
| | |
| Outcome assessors were NOT blind to the allocation of patients | 0 |
| Outcome assessors were blind to the allocation of patients | 1 |
| | |
| The timing of the measurement of the outcomes was NOT appropriate | 0 |
| The timing of the measurement of the outcomes was appropriate | 1 |
| | |
| Total number of points | /10 |


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METHODS

| | | | |
|---|---|---|---|
| Subject-Blinded | Yes ( ) | No ( ) | Unclear ( ) |
| Physician-Blinded | Yes ( ) | No ( ) | Unclear ( ) |
| Outcome Assessor-Blinded | Yes ( ) | No ( ) | Unclear ( ) |

PARTICIPANTS

| | |
|---|---|
| Number of eligible participants: ___ | Number enrolled in study: ___ |
| Number of males: ___ | Number of females: ___ |
| Number age 0 yo 3 months: ___ | Number age 3 to 6 months: ___ |
| Number greater than 6 months to 12 months: ___ | Number greater than 12 months: ___ |

INTERVENTION

| INTERVENTION | Drugs(specify)/Therapy | Dose | Duration | Rule |
|---|---|---|---|---|
| Treatment group 1: | | | | |
| Treatment group 2: | | | | |

COMMENT ON TREATMENT

| | | | |
|---|---|---|---|
| Withdraws: ___ | Yes ( ) | No ( ) | Unclear ( ) |

OUTCOMES

| OUTCOMES | Day in hospital (mean) | Death in hospital (n,%) | Long-term death (n,%) | Neurologic morbidity (specify) |
|---|---|---|---|---|
| Treatment group 1: | | | | |
| Treatment group 2: | | | | |


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| OUTCOMES | Change in mean airway pressure | Change in mean systemic pressure | Change in heart rate | Maximum methaemoglobin |
|---|---|---|---|---|
| Treatment group 1: | | | | |
| Treatment group 2: | | | | |

CHANGES IN PROTOCOL:

CONTACT WITH AUTHOR:

OTHER COMMENTS ON THIS STUDY:

## WHAT'S NEW

| Date | Event | Description |
|---|---|---|
| 3 July 2014 | New search has been performed | The search was originally run in 2004 (Bizzarro 2005). In this updated review we reran the search strategy until December 2013. |
| 3 July 2014 | New citation required but conclusions have not changed | We found three new studies. After we assessed the full articles for eligibility criteria we excluded these articles in (Checchia 2013; Kirbas 2012; Loukhanov 2011) our updated review. No new studies were included in the update. The conclusions were not changed.<br><br>One new author helped update this version: Fabiano T Barbosa.<br><br>We updated the methods section. We updated the risk of bias tool. We included a summary of findings table. |

## HISTORY

Protocol first published: Issue 4, 2004
Review first published: Issue 4, 2005


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| Date | Event | Description |
|---|---|---|
| 31 July 2005 | New citation required and conclusions have changed | Substantive amendment |

## CONTRIBUTIONS OF AUTHORS

Dr Matthew Bizzarro (MB), Dr Ian Gross (IG) and Fabian T Barbosa (FTB) updated this review.

Conceiving the review: MB
Co-ordinating the review: FTB
Undertaking manual searches: MB
Screening search results: MB, IG
Organizing retrieval of papers: MB
Screening retrieved papers against inclusion criteria: MB
Appraising quality of papers: MB, IG, FTB
Abstracting data from papers: MB
Writing to authors of papers for additional information: MB
Providing additional data about papers: MB
Obtaining and screening data on unpublished studies: MB
Data management for the review: MB
Entering data into Review Manager (RevMan 5.2): FTB
RevMan statistical data: MB, FTB
Other statistical analysis not using RevMan: MB
Double entry of data: MB, FTB
Interpretation of data: MB, IG
Statistical inferences: MB
Writing the review: FTB
Securing funding for the review: N/A
Performing previous work that was the foundation of the present study: MB
Guarantor for the review (one author): MB
Person responsible for reading and checking review before submission: MB, IG, FTB

## DECLARATIONS OF INTEREST

Matthew Bizzarro: none known

Ian Gross: none known

Fabiano T Barbos: none known

## SOURCES OF SUPPORT

### Internal sources

- Yale University School of Medicine, USA.

### External sources

- No sources of support supplied

## DIFFERENCES BETWEEN PROTOCOL AND REVIEW

February 2014

1. One new author was included (Fabiano T Barbosa).
2. We updated the methods section. Risk of bias tools were included in this updated review and items analysed before in 'risk of bias in included studies' (method of randomization, concealment of allocation, blinding of intervention, completeness of follow up, and blinded outcome assessment) were renamed to random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias.
3. Items of the risk of bias tables were reanalysed.


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4. We included a summary of findings table.

## NOTES

None

## INDEX TERMS

### Medical Subject Headings (MeSH)

Administration, Inhalation; Heart Defects, Congenital [*surgery]; Heart Diseases [congenital] [surgery]; Hypertension, Pulmonary [*drug therapy] [mortality]; Nitric Oxide [administration & dosage] [*therapeutic use]; Postoperative Complications [*drug therapy] [mortality]; Randomized Controlled Trials as Topic

### MeSH check words

Child; Child, Preschool; Humans; Infant


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